DATA 255, Spring 2026

Lab #3

Due by 3pm, April 29th, 2026

Notes

This lab assignment should be submitted on Canvas.

You may collaborate on lab but must **independently** write code/solutions. Copying others' codes or directly using LLM's generated codes will be treated as cheating and will not be tolerated and result in a **zero score** for the lab assignment (minimal penalty) or a failing grade for the course. Your work will be graded in terms of correctness, completeness, and clarity, not just the answer. Thus, correct answers with no or poorly written supporting steps may receive very little credit.

Please download 'L3_HR_train.zip', 'L3_LR_train.zip', 'L3_HR_train1.zip', 'L3_LR_train1.zip', 'L3_HR_train2.zip', 'L3_LR_train2.zip', 'L3_HR_train3.zip', 'L3_LR_train3.zip', 'L3_HR_train4.zip', 'L3_LR_train4.zip', 'L3_HR_valid.zip', and 'L3_LR_valid.zip' from canvas/Modules/Lab3.
Please build a deep neural network, train it, and run it on NPU according to the following instructions.

The goal of Lab2 was to improve PSNR.
The goal of Lab3 is to improve latency of the model.

Once you've confirmed that your model runs successfully and performs well on the validation set on NPU, please submit your model along with your training script on Canvas.
The calibration data for quantization is supposed to be a part of the training dataset.

For this assignment, please submit:
> Your training script. **Please include your model architecture (Pytorch) and the trained weight (xxx.pth).**
> Your .onnx to .mxq conversion script file
> Your xxx.mxq model file

Lab assistant or ISA will run your xxx.mxq file on the test dataset and assign your score (mean PSNR on NPU) on the leaderboard.
Please ensure beforehand that your model works after you compile it and that you're happy with the score.
Please ensure that the code is executable on NPU. If the code is not executable on NPU, your score will be zero.
Please also allow a day or two to post your score in the leaderboard.
You can resubmit your code as many times before the due date. But the score will be deducted as the following formula.

$$final\ score = score\ \times r^{t-1}$$
$$r: 0.95$$
$$t: number\ of\ trial\ to\ submit$$

Lab Instructions

Step 1. Build Deep Neural Network

Low-resolution (LR) images lose detail, gain noise, and artifacts from downscaling or degradation. Traditional upscaling (e.g., Bicubic Interpolation) fails to recover details, adds blur. Blurry images are unfortunately common and are a problem for medical imaging (e.g., missing MRI lesions) and surveillance (e.g., unclear license plates).
The objective of 2nd lab is to develop a Super-resolution model that restores fine details and mitigates noise in degraded low-resolution images.
Please build a Super-Resolution model from scratch using Pytorch.

NOTE:

You do not need to define activation function, loss function from scratch.
The input format (Image size) of the model is **[256, 256, 3].**
The output format (Image size) of the model is **[256, 256, 3].**
If your neural architecture is same as others, we treat this as cheating and result in zero score.

Step 2. Training

Please train the neural network with the training dataset.

NOTE:

You do not need to implement backward pass and gradient descent from scratch
You can use the following Pytorch backward pass.

```
# Backward pass
optimizer.zero_grad()  # clear the gradients of all optimized variables
loss.backward()
optimizer.step()  # Update parameters (gradient descent)
```

You might try various hyper parameters such as batch size, weight initialization method, learning rate, and number of epochs.
You can use Pytorch data loader.
You might need data augmentation to improve accuracy.

Step 3. Evaluate the trained model on NPU

Please convert(export) your Pytorch model to onnx file. Then, compile it to mxq file.
Please run the mxq file on NPU against the validation dataset and evaluate mean PNSR.

Grading criteria for lab2

Mean PSNR (dB) of test dataset on NPU	Mean Latency (ms) of test dataset on NPU	score
>25	<6	18
>25	<8	17
>25	<10	16
>25	<12	15
>25	<14	14
>25	<16	13
>25	<18	12
>25	<20	11
>25	>20	10
<=25	Any	8

NOTE:

The mean PSNR of valid dataset is 23.1dB. The mean PSNR of test dataset is 23.3dB. Your task is to improve the mean PSNR of valid and test datasets higher than 25dB using your model. Then, you need to optimize the model to minimize the latency. Only Validation dataset is given to the students. Lab assistants will evaluate your model using the test dataset on NPU.

The top 3 submissions with the lowest latency will be selected to present their work in class and will receive an additional 2 points.